

Lightning: A big static discharge

1. Weather systems happening inside clouds cause charged particles to move around. Inside the cloud there are different types of water particles such as lighter tiny ice crystals and heavier soft hail and slushy ice pellets
2. While charged particles are being moved around, they rub against each other becoming charged by friction. The heavier particles tend to gain negative charges and the lighter particles tend to become positive by losing negative charges.
3. This causes the charges inside the cloud to polarize into groups of negatives and positives with the heavier negative particles coming to the bottom of the cloud and the lighter positive particles coming to the top
4. The charges inside the cloud become so great, that they want to discharge to get back to neutral. The charges are also able to attract the charges on the ground below.
5. The strong force of attraction between the cloud and the ground causes electrons to travel down from negative areas of the cloud while the positives on the ground that they are attracted to move up as much as they can in their system. Where the negatives and the positives meet, a bolt of lightning is formed.
6. The lighting you see is moving electrons. The electrons are so charged and moving so fast that they heat up the air around them when moving; the heat is converted to light energy that is visible to us.